01 QUESTION ONE · MULTIMEDIA BASICS

Define Multimedia. What are the components of Multimedia? What are the different applications of Multimedia?

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DEFINITION OF MULTIMEDIA

The word **multimedia** is made of two parts — *multi*, meaning many, and *media*, meaning the ways in which information is presented. Together they mean the use of **more than one medium at the same time** in a single presentation. In computing, multimedia is the **integration of text, graphics, images, sound, video and animation** inside one digital system.

That last word, **interaction**, is what separates true multimedia from a film or a book. A typical multimedia application lets the user click, choose, type or drag, and the content responds in front of them.

Linear and Non-linear Multimedia

In **linear** multimedia the story runs in one fixed order from start to finish, like a slide show, so the viewer can only watch. In **non-linear** multimedia the user is given control and jumps to any branch he likes; almost all modern teaching and business software is non-linear.

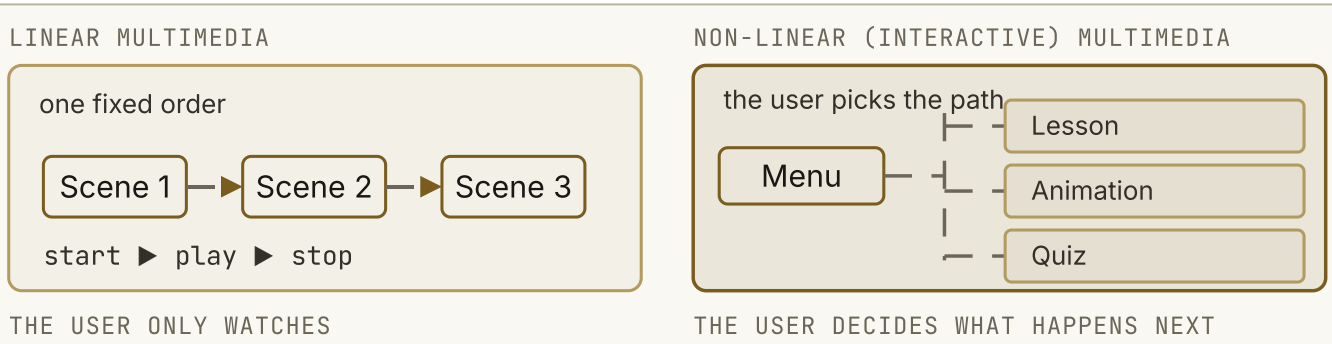


FIG. 1 — TWO KINDS OF MULTIMEDIA. LINEAR CONTENT MUST BE WATCHED IN THE ORDER IN WHICH IT WAS MADE; INTERACTIVE CONTENT HANDS THE CONTROL TO THE USER, WHO JUMPS FROM THE MENU TO WHATEVER INTERESTS THEM.

COMPONENTS OF MULTIMEDIA

Every multimedia system is built from six components, each doing one job.

How the Components Work Together

None of these components is useful alone. **Text** gives the facts and instructions, **graphics** and **images** give the picture, **sound** and **video** give the movement and the voice, and **animation** joins them into one controlled flow. A single piece of educational software may show a written lesson, an animated diagram of the heart, a recorded explanation and a short test at the end — and the user moves between them at will. Hardware makes this possible, because a computer can store and play sound and moving pictures as easily as it stores text.

COMPONENT	WORK IT DOES
Text	Gives facts, instructions and labels
Graphics	Draws lines, shapes and charts
Image	A single still photograph
Sound	Music, speech and effects
Video	Moving pictures with sound
Animation	Motion created by the computer

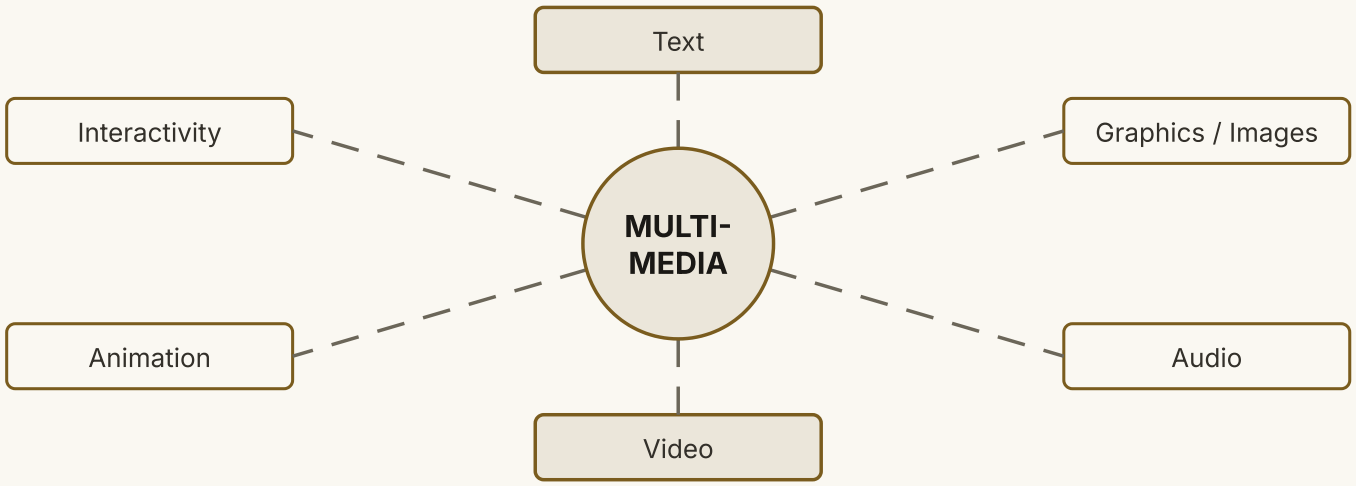


FIG. 2 — THE SIX COMPONENTS OF MULTIMEDIA. THE CENTRE IS NOT A SEVENTH MEDIUM; IT IS THE COMBINATION THAT TURNS SEPARATE MEDIA INTO ONE INTERACTIVE EXPERIENCE.

APPLICATIONS OF MULTIMEDIA

(a) Education: lectures become video, diagrams become animations and every topic can be replayed as often as the student likes.

(b) Entertainment: films carry picture and sound together, and games join all six components at once.

(c) Business and advertising: a product is shown, rotated, zoomed and compared before it is bought.

(d) Medicine: body parts are shown as 3-D models so a surgeon can practise before operating.

(e) Journalism: a live newsroom mixes recorded video, sound and text into one report.

(f) Engineering: a design is walked through as a virtual walkthrough before it is built.

(g) Government: citizens learn and apply for services through a single screen.

(h) Virtual reality and gaming: the user walks inside a computer-built world and every movement is answered instantly.

From these eight uses one conclusion follows. Multimedia earns its place only where the subject itself is better understood when it can be **seen and heard**. Where a list of names or a table of figures would do just as well, adding sound and motion only wastes time and confuses the message. That is why the same subject may need no multimedia at all in one lesson and a great deal of it in the next.

HTML EXAMPLE

TEXT, IMAGE, AUDIO AND VIDEO TOGETHER IN ONE PAGE — THE WHOLE IDEA OF MULTIMEDIA IN A SINGLE FILE.

THE SAME FILE WITHOUT A VIDEO TAG STILL WORKS ON A COMPUTER THAT CANNOT PLAY IT, BECAUSE THE `CONTROLS` ATTRIBUTE GIVES THE USER A PLAY AND PAUSE BUTTON AND THE `WIDTH` KEEPS THE PICTURE INSIDE THE PAGE. THE `SOURCE` TAG NAMES THE FILE AND ITS TYPE, SO THE BROWSER KNOWS HOW TO PLAY IT.

MULTIMEDIA.HTML

```
<!DOCTYPE html>
<html>
<head><title>My Multimedia Page</title></head>
<body>
  <h1>Welcome</h1>
  <p>This page uses several media together.</p>
  <img src="college.jpg" alt="College photo" width="320">
  <audio controls>
    <source src="song.mp3" type="audio/mpeg">
  </audio>
  <video controls width="320">
    <source src="clip.mp4" type="video/mp4">
  </video>
</body>
</html>
```

CONCLUSION

Multimedia is not one medium but several working together, under the user's control.

02 QUESTION TWO · IMAGE FILE FORMATS

Discuss the image file formats and explain the differences between them.

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PYTHON PROGRAM ON PAGE 4

INTRODUCTION TO IMAGE FILE FORMATS

An **image file format** is the standard way a picture is written on disk. The format decides how colours are stored, whether the file is compressed, whether transparency is possible and which programs can open it. Choosing wrongly costs a great deal: the same picture may be 40 KB in one format and 4 MB in another.

1. JPEG (JOINT PHOTOGRAPHIC EXPERTS GROUP)

JPEG is a **lossy** format built for photographs. It splits the picture into 8×8 blocks, converts each block into frequencies and deletes the weak ones, so the file is small but the fine detail is softened. It does not support transparency.

2. PNG (PORTABLE NETWORK GRAPHICS)

PNG is **lossless**: nothing is thrown away, so the picture stays exact. It supports a transparent background, which is why it is the normal choice for logos, icons and screenshots, at the cost of a larger file than JPEG for photographs.

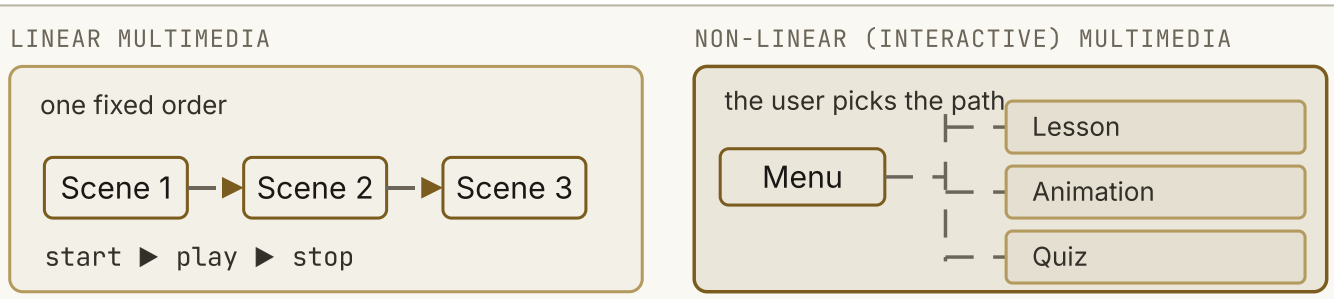


FIG. 1 — TWO KINDS OF MULTIMEDIA. LINEAR CONTENT MUST BE WATCHED IN THE ORDER IN WHICH IT WAS MADE; INTERACTIVE CONTENT HANDS THE CONTROL TO THE USER, WHO JUMPS FROM THE MENU TO WHATEVER INTERESTS THEM.

3. GIF is the oldest web format. It holds only 256 colours but supports simple animation and transparency.

4. BMP is an uncompressed Windows bitmap — perfect quality, very large files, used only inside Windows.

5. TIFF is lossless, very high quality and supports many pages, so printers and scanners use it.

6. WebP is Google's modern format. It is both lossy and lossless and is roughly 25–35 % smaller than JPEG or PNG at the same quality.

7. SVG is a **vector** format: it stores lines and shapes, so a logo can be enlarged to any size with no loss.

8. RAW is exactly what the camera sensor recorded, with almost no processing, so a photographer can adjust the picture fully later. The files are the largest of all and used only in professional work.

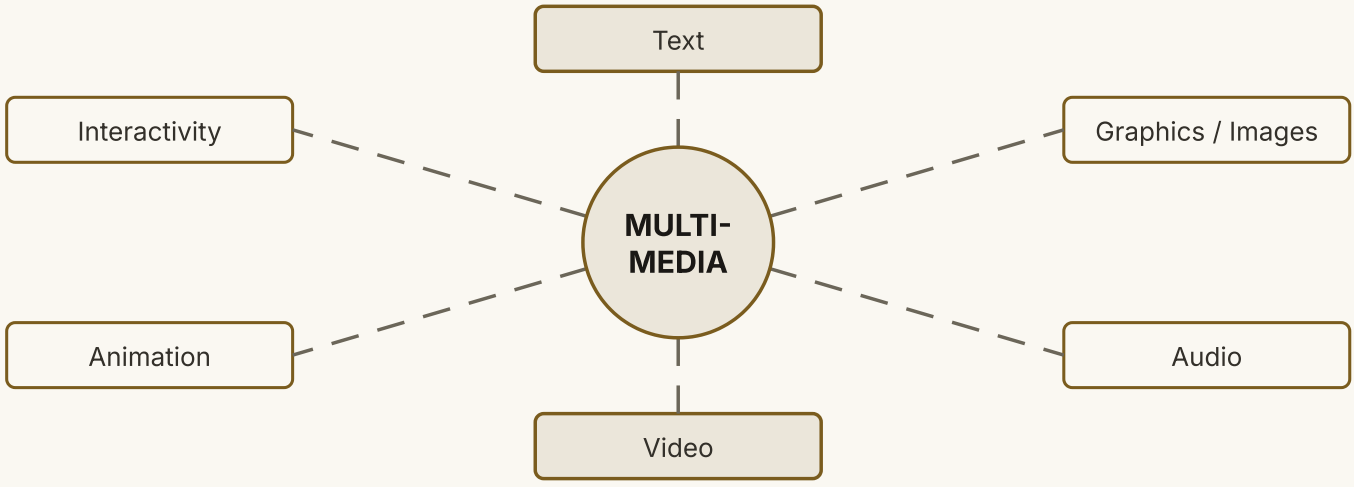


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DIFFERENCES BETWEEN THE IMAGE FORMATS

Formats are compared on two separate axes. The first asks **how the picture is built**: a **raster** image is a grid of coloured dots and goes blocky when enlarged, while a **vector** image is made of lines and shapes and stays sharp at any size. The second asks **how the file is squeezed**: **lossy** compression throws detail away to save space, while **lossless** compression only squeezes the pattern and keeps every pixel.

FORMAT	TYPE	BEST FOR
JPEG	Lossy	Photographs
PNG	Lossless	Logos, transparency
GIF	Lossless, 256 colours	Simple animation
SVG	Vector	Drawings and icons

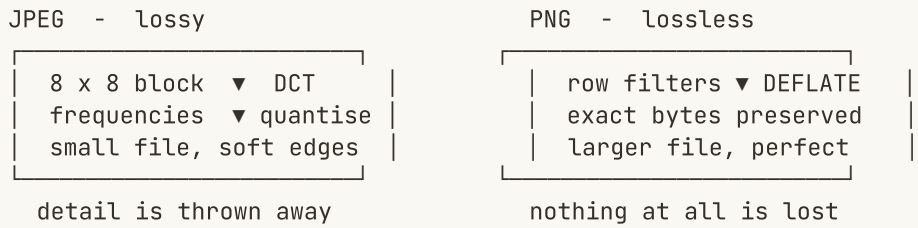


FIG. 1 — THE TWO COMPRESSION IDEAS SIDE BY SIDE, IN TEXT FORM. JPEG TURNS EACH SMALL BLOCK OF PIXELS INTO FREQUENCIES AND THEN DELETES THE WEAK ONES, WHICH IS WHY IT IS SMALL; PNG FILTERS EACH ROW OF PIXELS AND THEN APPLIES A LOSSLESS DICTIONARY, WHICH IS WHY IT IS EXACT.

PYTHON PROGRAM

ONE PICTURE SAVED IN THREE FORMATS — USING THE PILLOW (PIL) LIBRARY. `IMAGE.OPEN` READS THE PICTURE ONCE AND KEEPS IT IN MEMORY, SO THE SAME IMAGE CAN BE WRITTEN OUT SEVERAL TIMES WITHOUT BEING LOADED AGAIN. THE `QUALITY` ARGUMENT IS THE ONLY SETTING THAT MATTERS FOR JPEG: A LOWER NUMBER GIVES A MUCH SMALLER FILE AND A VISIBLY SOFTER PICTURE. PNG AND BMP TAKE NO SUCH ARGUMENT, BECAUSE NEITHER THROWS ANYTHING AWAY. THE LAST LOOP SIMPLY PRINTS THE NAME OF EVERY FILE THAT HAS NOW BEEN WRITTEN TO THE FOLDER.

FORMATS.PY

```python
# store one picture in three different formats
from PIL import Image

img = Image.open("photo.jpg")
print("original :", img.size, img.mode)

img.save("copy.jpg", "JPEG", quality=75)   # lossy, small
img.save("copy.png", "PNG")                # lossless, exact
img.save("copy.bmp", "BMP")                # uncompressed, huge

for name in ("copy.jpg", "copy.png", "copy.bmp"):
    print(name, "saved")
```

Output

OUTPUT

```
original : (640, 480) RGB
copy.jpg saved
copy.png saved
copy.bmp saved
```

CONCLUSION

The right format is decided by the job, never by fashion.

What is an animation?

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DEFINITION OF ANIMATION

Animation is the technique of creating an illusion of movement by showing a series of still pictures one after another very quickly. Each picture is slightly different from the one before it, and when they arrive fast enough the human eye joins them into one smooth motion. Animation is therefore not one moving picture — it is many still pictures played at speed.

The reason this works is a property of human vision called the **persistence of vision**. When the eye sees a still image, the image stays in the brain for about one-tenth of a second after it has disappeared. If the next image arrives before that time runs out, the brain does not notice the gap and reads the whole set as continuous movement.

This also explains why an animation must be shown at a steady speed. If the frames arrive too slowly, each one is seen separately and the illusion breaks. If the frames arrive at an uneven rate, the motion looks jerky even though the pictures are correct.

How Animation is Created

An animator first plans the **storyboard** — a rough sketch of every important moment — then draws the **keyframes** that hold those moments and lets the computer fill in the **in-betweens** between them. The finished frames are then stored in order and played at a chosen **frame rate**, measured in frames per second (fps).

TYPES OF ANIMATION

Traditional 2-D animation is drawn frame by frame by hand on transparent celluloid, so an animator may produce thousands of drawings for a few minutes of film.

Computer 2-D animation is drawn once on a screen and then moved by the computer, so only the moving parts have to be redrawn and the same material can be reused many times.

3-D CGI builds a complete model in three dimensions and renders a fresh frame from it for every moment, which is why the camera can move freely around the object.

Stop-motion photography photographs real objects that have been moved a very small amount each time, and the movement appears when the pictures are joined.

Motion graphics animation works on text, logos and shapes, and **procedural** animation is produced entirely by code, with no artist drawing anything at all.

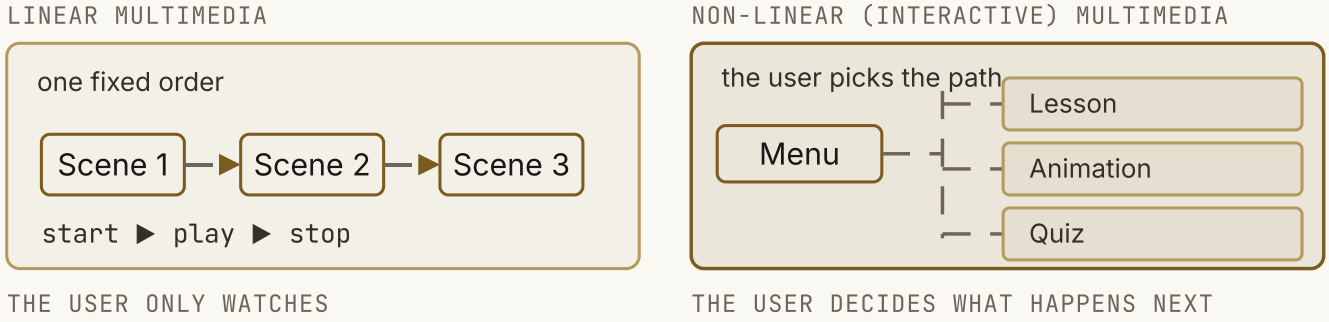


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KEY TERMS RELATED TO ANIMATION

Frame: one single still picture of the series.
Frame rate: how many frames are shown each second; film uses 24 fps.
Key frame: a frame that the artist draws by hand and that fixes the position of the object at one moment.
Storyboard: the rough sketch of the whole sequence, made before any drawing begins.
Sprite: one small image that is moved around a background.

USES AND APPLICATIONS OF ANIMATION

Animation is used in films and cartoons, in video games, in advertisements and in educational software, and it is the base of virtual reality, where the user moves inside a computer-built world and the picture answers every movement at once.

Animation earns its keep in teaching because it shows a change that a still picture cannot show. A plant growing, a current flowing, a cylinder turning into a cone and a light ray bending — each of these is understood in seconds when it moves, and may take a whole paragraph when it is only described. This is why animation is used to explain physics, mathematics, chemistry and the working of machines.

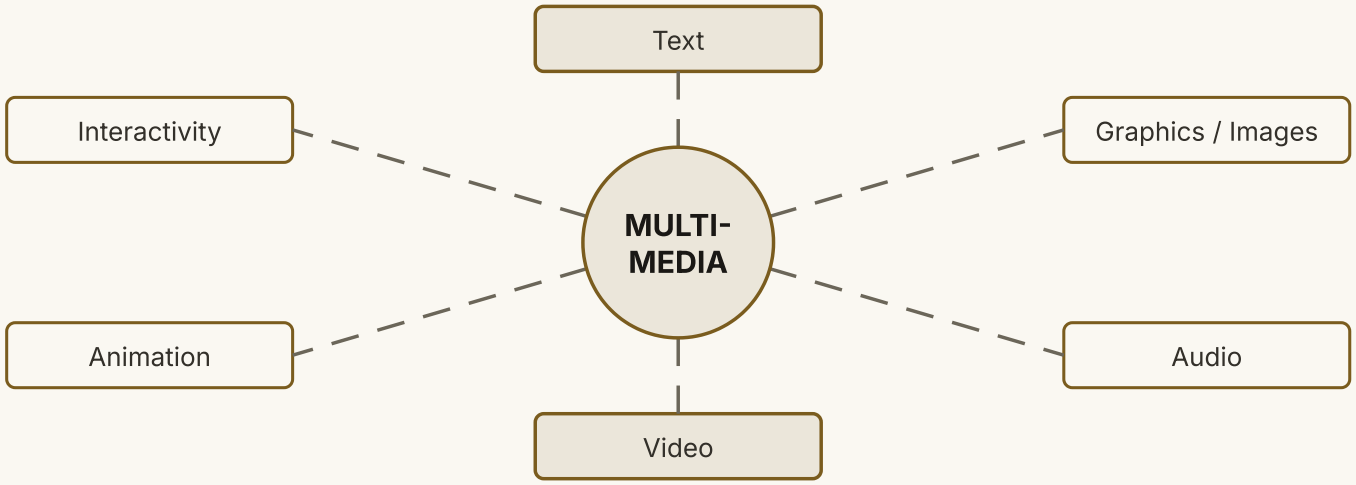


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CSS ANIMATION EXAMPLE

A MOVING SQUARE MADE WITH `@KEYFRAMES` AND THE `ANIMATION` PROPERTY.

MOVE.HTML

```
<!DOCTYPE html>
<html>
<head>
<style>
  @keyframes slide {
    from { left: 0px;   background: red; }
    to   { left: 260px; background: blue; }
  }
  .box {
    width: 80px; height: 80px; position: relative;
    animation: slide 3s linear infinite;
  }
</style>
</head>
<body>
  <div class="box"></div>
</body>
</html>
```

How the Example Works

`@keyframes slide` names the movement and describes its starting state with `from` and its ending state with `to`. The `.box` rule then plays it: `animation: slide 3s linear infinite` means run the movement once every three seconds, at an even speed, and never stop. Because the box is placed with `position: relative`, the `left` value in the keyframes moves it away from the left edge of the page.